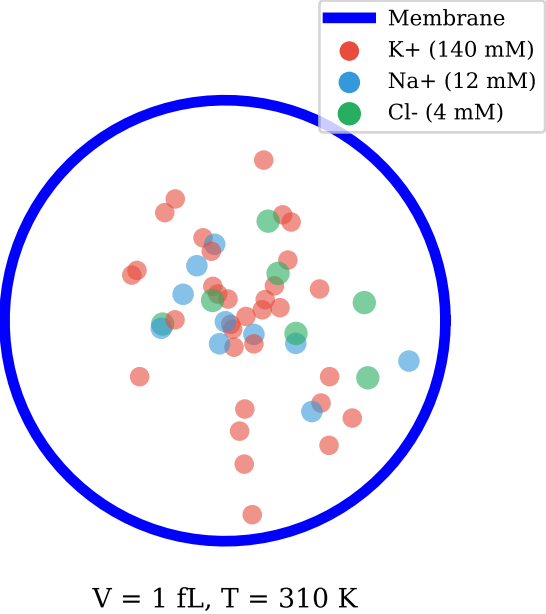
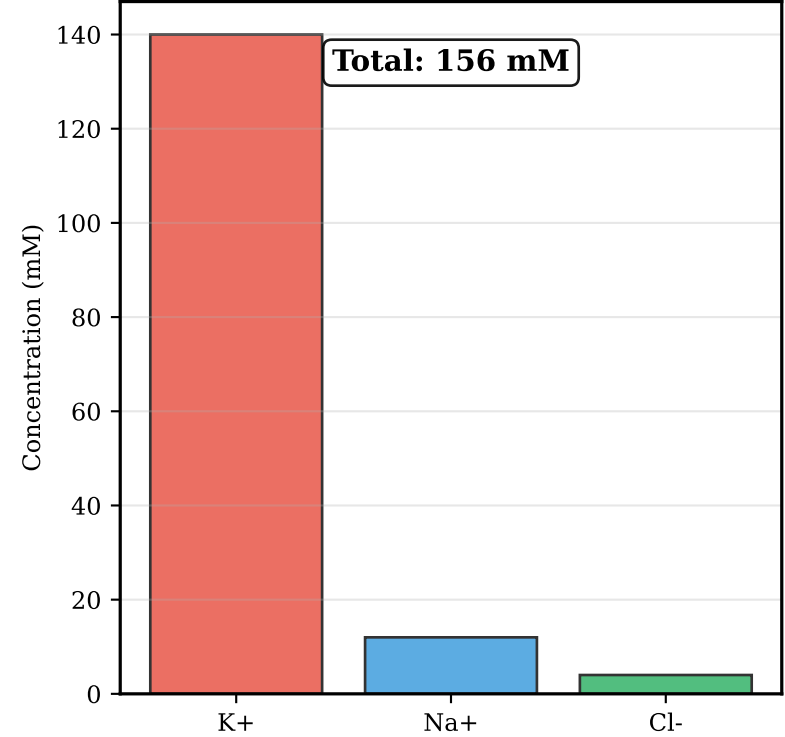


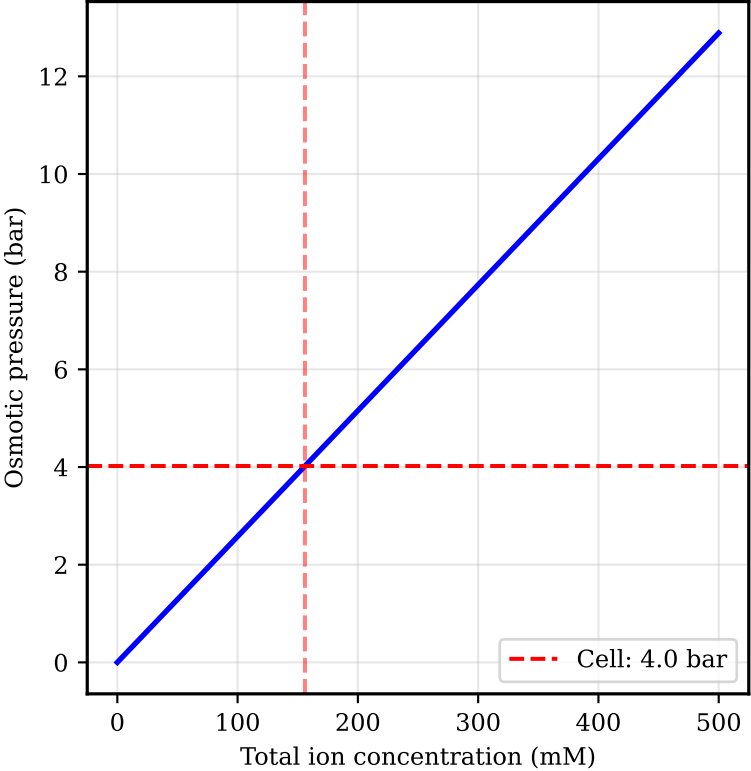
(A) Cellular Ion Chamber



(B) Intracellular Ion Concentrations



(C) Osmotic Pressure: $\Pi V = nRT$



(D) Categorical Ion Framework

CATEGORICAL CELLULAR FRAMEWORK

Container: Plasma membrane

- Volume V ~ 1 pL = 10^{-15} L
- Bounded, oscillatory boundary

Particles: Intracellular ions

- K+, Na+, Cl-, Mg2+, etc.
- N ~ 10^8 particles per cell

Categorical Structure:

- Ion channels: discrete transitions
- Binding sites: categorical states
- Compartments: partition boundaries

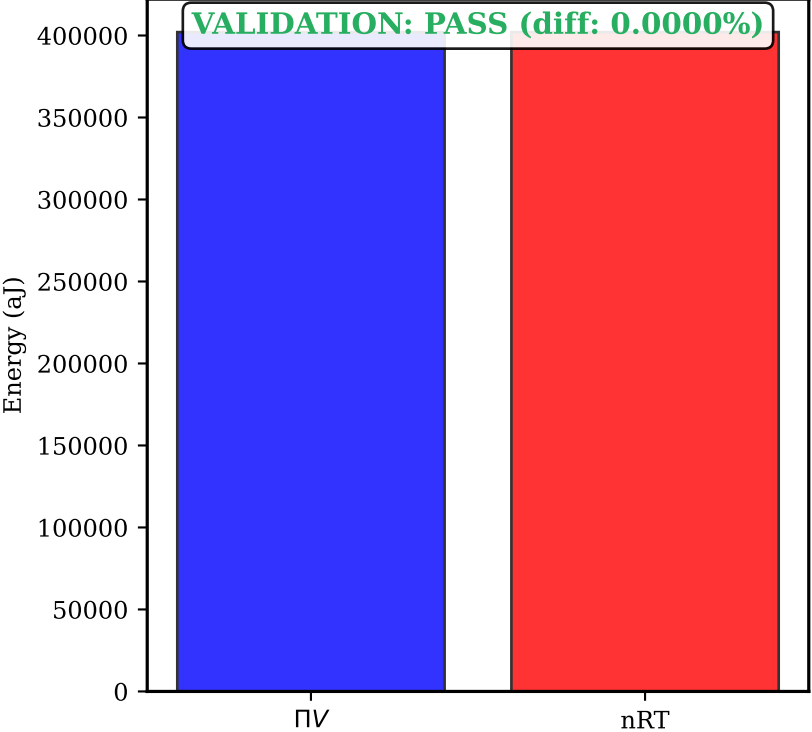
Temperature:

- T = 310 K (body temperature)
- T = $\hbar \text{bar}(\text{dM}/\text{dt})/k_B$ for ion flux

Pressure:

- Osmotic: $\Pi = c_{\text{eff}} * R * T$
- Categorical: $P = k_B T(\text{dM}/\text{dV})$

(E) Osmotic: $\Pi V = nRT$



(F) Cellular Application Summary

CELLULAR ION VALIDATION

Parameters:

- Cell volume: 1 fL
- Temperature: 310 K
- [K+]: 140 mM
- [Na+]: 12 mM
- [Cl-]: 4 mM
- Total: 156 mM

Calculations:

- N total: 9.39×10^{10} particles
- Osmotic pressure: 4.02 bar

Ideal Gas Law:

- $\Pi * V = 402065.0400$ aJ
- nRT = 402065.0400 aJ
- Difference: 0.000000%

VALIDATION: PASS

CONCLUSION: Intracellular ions obey ideal gas / Van't Hoff equation. The categorical framework applies to cellular thermodynamics.